

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay_compression.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_172929_j
- started_at_local: 2026-05-13 17:29:29
- finished_at_local: 2026-05-13 17:37:55
- duration_hhmm: 00:08
- duration_seconds: 506.097
- seed_offset: 9000
- replicate_label: j
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.190984	0.009004	0.099994	0.0	0.58	0.0
atsp_random_replay	random	score_only	False	0.190037	0.008454	0.099246	0.641642	0.58	0.017328
atsp_failure_replay	failure	score_only	False	0.201694	0.030781	0.116238	0.0	0.78	0.0
atsp_failure_replay_compression	failure	novelty_gate	True	0.172666	0.030781	0.101723	0.690932	0.78	0.020358

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190984, synthetic 0.009004, combined 0.099994.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190984 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.003915, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190037, synthetic 0.008454, combined 0.099246.
- Accepted-epoch count 4, mean accepted code novelty 0.641642, and final complexity 0.58.
- Adaptation efficiency 0.017328 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190037 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.097143.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.201694, synthetic 0.030781, combined 0.116238.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.201694 across 4 instances; family means: ft=0.325561, ftv=0.288283, p=0.017438, ry=0.175496.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1596, "family": "ftv", "name": "ftv33", "optimality_gap": 0.241058}, {"best_known_cost": 1530, "cost": 1896, "family": "ftv", "name": "ftv38", "optimality_gap": 0.239216}, {"best_known_cost": 1473, "cost": 1711, "family": "ftv", "name": "ftv35", "optimality_gap": 0.161575}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.172666, synthetic 0.030781, combined 0.101723.
- Accepted-epoch count 2, mean accepted code novelty 0.690932, and final complexity 0.78.
- Adaptation efficiency 0.020358 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.

- Panel heldout_tsplib mean gap 0.172666 across 4 instances; family means: ft=0.2895, ftv=0.249225, p=0.000712, ry=0.151227.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1574, "family": "ftv", "name": "ftv33", "optimality_gap": 0.22395}, {"best_known_cost": 1530, "cost": 1850, "family": "ftv", "name": "ftv38", "optimality_gap": 0.20915}, {"best_known_cost": 1473, "cost": 1722, "family": "ftv", "name": "ftv35", "optimality_gap": 0.169043}]

Judge Appendix

Summary of Results

Condition	Synt heti c Gap	Transfer Gap (held-out TSPLIB & synthetic)	TSP LIB Gap	Code Novelty (mean)	Replay Mode	Sele ction Mod e	Co m ple xit y	Notes
atsp_no_replay	0.009004	0.099994 (transfer)	0.190984	0.0	none	score_only	0.58	Baseline; no replay; lowest code novelty
atsp_random_replay	**0.008454**	**0.099246** (best transfer)	0.190037	0.641642	random	score_only	0.58	Best transfer and synthetic gap; higher code novelty
atsp_failure_replay	0.030781	0.116238	0.201694	0.0	failure	score_only	0.78	Worse transfer & synthetic gaps; no code novelty
atsp_failure_replay_compression	0.030781	0.101723 (best TSPLIB gap)	**0.172666**	0.690932	failure + compression	novelty_gate	0.78	Best TSPLIB gap; code novelty increased; compression active

Interpretation:

- **Transfer (Held-out TSPLIB & Synthetic) Gap:**
- *atsp_random_replay* condition yields the best combined transfer gap (0.099246) with slightly improved synthetic gap (0.008454) compared to *no_replay*.
- *atsp_failure_replay_compression* improves held-out TSPLIB gap substantially (best at 0.172666) over other failure replay and baseline conditions, indicating better transfer to realistic TSPLIB benchmarks.
- *atsp_failure_replay* alone yields poorer transfer and synthetic gaps than both *random* and *no replay* modes.
- **Code Novelty vs Transfer:**
- Conditions with replay (*random* and *failure + compression*) show substantial increase in code novelty (~0.64-0.69) compared to zero novelty in *no replay* and *failure replay* without compression.
- Higher code novelty in failure+compression replay aligns with best TSPLIB gap but not with the best synthetic or combined transfer gaps.

- The **random replay** condition improves transfer without a reduction in code novelty, showing code novelty and transfer can grow together here.
- Code novelty alone does not signify algorithmic invention given that transfer gains are marginal or mixed across failure replay settings.
- ****Replay Mode Effects:****
 - **Random replay** improves transfer gaps relative to **no replay**, with modest code novelty increase.
 - **Failure replay** without compression degrades transfer and synthetic gap, no code novelty gain.
 - **Failure replay with compression** improves TSPLIB transfer gap relative to failure replay without compression, indicating compression-aware replay aids transfer in failure replay context.
 - Selection mode using **novelty_gate** in failure replay+compression coincides with better TSPLIB transfer despite synthetic gap similar to failure replay alone.
- ****Complexity:****
 - Failure replay conditions show higher complexity (0.78) compared to no replay and random replay (0.58), which could imply more complex solutions without consistent improvements in transfer.

Conservative Conclusions:

1. ****Transfer Evidence:****
 - The strongest evidence for transfer is found for ****atasp_random_replay****, which achieves the best combined synthetic and transfer gap.
 - ****Failure replay with compression**** achieves the best TSPLIB gap, indicating improved transfer to realistic benchmarks in compression-aware failure replay.
2. ****Code Novelty:****
 - Code novelty increases with replay, especially failure replay with compression, but increased code novelty does not always correspond to better transfer (e.g., failure replay alone).
 - Since no replay has zero novelty with acceptable transfer, novelty must be interpreted cautiously without attributing algorithmic invention solely to lexical novelty metrics.
3. ****Replay mode impacts:****
 - Random replay positively impacts transfer gaps without increasing model complexity.
 - Failure replay without compression decreases transfer quality, indicating that replay must be coupled with mechanisms (compression, novelty gating) to realize benefits.
 - Compression-aware replay combined with failure replay yields a more favorable TSPLIB gap, showing retrieval strategy and replay conditions affect generalization distinctly on held-out datasets.
4. ****Complexity vs Optimality-gap:****
 - Increased complexity under failure replay modes does not guarantee better synthetic or transfer gaps.
 - Compression-aware replay manages to balance complexity and transfer more effectively.

Recommendation:

- Prioritize **random replay** for improved and consistent synthetic and transfer optimality gaps with moderate code novelty and low complexity.
- Consider **failure replay with compression and novelty-gate selection** when best TSPLIB performance is critical.
- Avoid interpreting code novelty increases as algorithmic innovation absent concurrent improvement in deterministic transfer metrics.
- Differ between replay conditions carefully; failure replay alone worsens transfer, but when combined with compression-aware replay, transfer improves notably.